

Shiyu Yue

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EDUCATION

Sun Yat-sen University, Guangdong, P.R.China
Bachelor in Physics, School of Physics and Astronomy
- averaged GPA : 3.9 / 4.0

Sept. 2020 – Present

PUBLICATIONS

Tang, H., **Yue, S.** et al., 2023, the 35th URSI General Assembly and Scientific Symposium on Radio Science, *ArXiv:2307.03453* (co-first author)

Tang, H., **Yue, S.** et al., “May I trust you?”: eXplainable AI for Radio Galaxy Morphology Identification *in prep.* (co-first author)

RESEARCH EXPERIENCE

A Fast Statistical Algorithm for Cosmology Based on Multiscale Analysis and Application

Oct. 2021 - Present

Cosmology & Cosmic Statistics

Advisor: Prof. Longlong Feng, Sun Yet-sen University ; Bachelor Thesis

- Utilized C++ Programming to develop a highly efficient algorithm for calculating the two-point correlation function (2PCF) **with a complexity of $O(N)$** , where N represents the number of grids. Compared this approach to traditional count-in-cell methods, demonstrating the consistency between these two methods numerically.
- Applied the algorithm to measure the **three-point correlation function (3PCF)** using Dark Matter realization obtained from the MDPL2 simulation, **involving more than 10^8 particles**. Analyzed and compared the results with their theoretical counterparts, employing Python programming. Accomplished the entire analysis within a remarkable time frame of less than 8 hours.
- Evaluated and determined the optimal set of parameters for the algorithm across different number densities. Successfully applied the algorithm to Halo realizations from the Quijote simulation.
- Ongoing work: Utilizing Markov Chain Monte Carlo (MCMC) techniques to evaluate the assembly bias and constrain cosmological parameters from the 3PCF, specifically considering different triangle shapes. (paper in prep.)

Explainable Artificial Intelligence Applications on Radio Galaxy Morphology Classification

Jul. 2021 - Aug. 2021 / Sept. 2022 - Present

Radio Astronomy & Machine Learning

Advisor: Dr. Hongming Tang; Tsinghua University

- Developed a Convolutional Neural Network(CNN) model for radio galaxy classification.
- Employed Local Interpretable Model-agnostic Explanations (LIME) for **model behavior analysis** and **enhanced LIME visualization** for better interpretability.
- Evaluated the impact of hyper-parameter combinations of LIME and segmentation methods and tested LIME stability among the dataset. (paper in prep.)
- Ongoing work: Apply evaluation metrics like deletion/insertion curve to quantify LIME performance .

Planetary Nebula

Advisor: Prof. Quentin Parker; Hong Kong University

- Determined **Central Star of Round Galactic Planetary Nebula(CSPN) in HASH database** using observation like PanSTARRS, DECaPS DR2 and recorded coordinates.
- Evaluated the statistics of **offset between CSs and the centre of PN**.
- Compared the coordinates results with those found by GAIA with two different automatic methods.
- Presented a talk on Annual Research Jamboree of Laboratory for Space Research and awarded the *Best Speaker*
- Ongoing work: utilize a combination of HASH and an automated search method in GAIA as dataset and apply Artificial Intelligence to the hunt for CSPN.

SKILLS

Computer Skills:

- Proficient with Python and C++.
- Familiar with Machine Learning (Deep Learning Framework: PyTorch).
- Familiar with Linux.

English level:

- IELTS scores 7.5

ADWARDS

First award of scholarship, Sun Yet-sen University	Dec. 2023
Outstanding Campers, Summer Camp of Astronomy, Nanjing University	Jul. 2023
First award of scholarship, Sun Yet-sen University	Dec. 2022
First award of scholarship, Sun Yet-sen University	Dec. 2021
China Aerospace Foundation-Uniasia Aerospace scholarship, Sun Yet-sen University	Nov. 2021
Second award, South China Astronomical Contest	Nov. 2020

CONFERENCE & WORKSHOPS

International Workshop on Machine Learning in Astronomy, Hybird (Invited Talk)	Nov. 2023
International Workshop on Intelligent Computing in Astronomy, CHINA (Talk*)	Nov. 2023
Workshop on Cosmological Simulation and Statistics for DESI, Euclid Sky Survey, CHINA	Jul. 2023
2022 UFITs Cosmology Summer School, CHINA	Aug. 2022

OUTREACH

Student Research Assistant, Laboratory for Space Research, Hong Kong University.	Sept. 2023 - Dec. 2023
Chairman, Media Center of School of Physics and Astronomy, Sun Yat-sen University.	Sept. 2022 - Sept. 2023
Propaganda Department Manager, Astronomical Association, Sun Yat-sen University.	Sept. 2022 - Sept. 2023
Chairman, Astronomical Association, Guangzhou No.2 High School.	Sept. 2018 - Sept. 2019

* presented by co-first author